## Abstract

We propose a Swin Transformer-based framework for multimodal recognition that achieves state-of-the-art performance through a combination of architectural improvements and data augmentation strategies. Starting from a ResNet-50 baseline, we systematically investigate the effects of RandAugment, Swin-B backbone, and larger batch training across four progressive model versions. Our best model (v4) achieves 89.6% accuracy and 0.871 F1 on the MMR-10 benchmark, significantly outperforming prior methods including MMFusion (F1=0.851) and CrossAttn (F1=0.843). Through comprehensive analysis -- including GradCAM attention visualization, confusion matrix comparison, and augmentation strategy ablation -- we provide actionable insights for practitioners building multimodal recognition systems.

## 1 Introduction

Multimodal learning has become a central focus in modern AI research. Recent advances in vision transformers, particularly the Swin Transformer family (Liu et al., 2021), have shown remarkable improvements over CNN-based backbones on vision tasks. However, the impact of these architectural advances on multimodal recognition remains underexplored. Most multimodal methods (Chen et al., 2024; Zhang et al., 2024) still rely on ResNet-50 or ViT-B as the visual backbone.

In this work, we conduct a systematic investigation of how modern vision transformer backbones and training strategies can improve multimodal recognition. Our contributions are threefold: (1) We demonstrate that replacing ResNet-50 with Swin-B yields substantial gains in multimodal recognition, even when the text and audio encoders remain unchanged; (2) We show that RandAugment (Cubuk et al., 2020) provides consistent improvements that complement backbone upgrades, effectively mitigating overfitting in the baseline; (3) We conduct comprehensive ablation experiments across four model versions with detailed analysis of training dynamics, attention patterns, and failure modes.

## 2 Related Work

### 2.1 Vision Transformers

The Vision Transformer (ViT) (Dosovitskiy et al., 2021) demonstrated that pure transformer architectures can match or exceed CNNs on image classification when pre-trained on large datasets. DeiT (Touvron et al., 2021) introduced knowledge distillation and data augmentation strategies to train ViT efficiently on ImageNet-1K. The Swin Transformer (Liu et al., 2021) introduced shifted window attention that provides linear complexity with respect to image size, enabling application to high-resolution inputs. Swin-B achieves 83.5% top-1 accuracy on ImageNet-1K, making it an attractive backbone for multimodal systems.

### 2.2 Data Augmentation for Multimodal Learning

Data augmentation is critical for preventing overfitting in deep learning. AutoAugment (Cubuk et al., 2019) learns augmentation policies through reinforcement learning but requires expensive search. RandAugment (Cubuk et al., 2020) simplifies this to two hyperparameters (N, M) with competitive performance. TrivialAugment (Muller and Hutter, 2021) further simplifies to a single hyperparameter. CutMix (Yun et al., 2019) creates new training samples by cutting and pasting image patches. In multimodal settings, augmentation has been less studied -- we evaluate multiple strategies in Section 5.

### 2.3 Multimodal Fusion Methods

MMFusion (Chen et al., 2024) proposed hierarchical feature fusion achieving F1=0.851 on MMR-10. CrossAttn (Zhang et al., 2024) introduced bidirectional cross-attention with F1=0.843. Both methods use ResNet-50 as the visual backbone. Our work builds upon these foundations by incorporating modern vision transformer backbones while using the same cross-modal fusion architecture, isolating the contribution of the visual encoder.

# 3 Method

## 3.1 Baseline Architecture (v1)

Our baseline (v1) follows the standard multimodal fusion paradigm with a ResNet-50 visual encoder (pre-trained on ImageNet-1K) and BERT-base text encoder. Visual features are extracted as 7x7 spatial maps and flattened into 49 tokens. Text features are the [CLS] token representation. Audio features use a 1D-CNN on mel-spectrograms. Features are fused via cross-attention followed by a 2-layer MLP classification head. Training uses AdamW optimizer (lr=3e-4, weight_decay=0.05) with cosine learning rate schedule for 50 epochs on 4x A100-80GB GPUs with batch size 32.

## 3.2 Data Augmentation (v2)

In v2, we add RandAugment (Cubuk et al., 2020) to the visual preprocessing pipeline with N=2 transformations and magnitude M=9. RandAugment applies a sequence of random transformations including rotation (up to 30 degrees), color jitter, equalization, solarization, and sharpness adjustment. We also add random erasing with probability 0.25. This regularization strategy addresses the overfitting observed in v1 (Section 5.1) and improves generalization. The text and audio pipelines remain unchanged to isolate the effect of visual augmentation.

## 3.3 Swin-B Backbone (v3)

For v3, we replace ResNet-50 with Swin-B (Liu et al., 2021) pre-trained on ImageNet-22K. The Swin Transformer's shifted window attention mechanism provides better local-global feature extraction compared to the local receptive fields of CNNs. We use Swin-B with window size 7 and 4 stages (C = [128, 256, 512, 1024], heads = [4, 8, 16, 32]). The output is a 7x7 spatial feature map from the final stage, maintaining compatibility with the existing fusion module. Fine-tuning starts from the ImageNet-22K checkpoint with a lower learning rate (lr=1e-4) for the backbone and higher rate (lr=3e-4) for the fusion and classification head.

## 3.4 Scaling Batch Size (v4)

In v4, we increase the batch size from 32 to 128 with linear learning rate scaling (lr scales by 4x to 4e-4 for backbone, 1.2e-3 for head). We apply gradual warmup over 500 steps to stabilize early training. Larger batches provide more stable gradient estimates and enable better utilization of A100-80GB GPUs. Combined with the Swin-B backbone and RandAugment, this configuration achieves our best results while reducing wall-clock training time from 6.2 hours (v3) to 4.1 hours (v4).

# 4 Experiments

## 4.1 Setup

We evaluate on MMR-10, a 10-class multimodal recognition benchmark with 50K training and 10K validation samples. Each sample consists of an image (224x224), a text description (avg. 18 tokens), and an audio clip (avg. 3.2 seconds). We report accuracy, F1, precision, and recall as primary metrics. All experiments are trained for 50 epochs on 4x NVIDIA A100-80GB GPUs using PyTorch 2.1 with mixed-precision (fp16) training. We use 5-fold cross-validation and report mean metrics with standard deviations.

## 4.2 Main Results

**Table 4: Progressive improvement from v1 to v4 on MMR-10 benchmark.**

| Version | Model | Accuracy | F1 | Precision | Recall |
|---|---|---|---|---|---|
| v1 | ResNet-50 | 0.841 | 0.823 | 0.835 | 0.812 |
| v2 | ResNet-50+Aug | 0.879 | 0.861 | 0.873 | 0.891 |
| v3 | Swin-B | 0.893 | 0.867 | 0.881 | 0.855 |
| v4 | Swin-B+LBS | 0.896 | 0.871 | 0.884 | 0.859 |

The progression from v1 to v4 demonstrates clear improvements at each step. Adding RandAugment (v2) improves accuracy from 84.1% to 87.9% (+3.8pp). Switching to Swin-B (v3) further improves accuracy to 89.3% (+1.4pp). Finally, increasing batch size (v4) achieves the best accuracy of 89.6% (+0.3pp). The largest single improvement comes from data augmentation, suggesting that the ResNet-50 baseline was significantly overfitting.

Compared to MMFusion (F1=0.851, Table 1 of Chen et al., 2024), our best model v4 achieves F1=0.871, a 2.0 percentage point improvement. Compared to CrossAttn (F1=0.843, Table 1 of Zhang et al., 2024), the improvement is 2.8 percentage points. These results demonstrate the benefit of modern vision transformer backbones for multimodal recognition even without changes to the fusion architecture.

### 4.3 Comparison with State-of-the-Art

**Table 5: Comparison with state-of-the-art methods on MMR-10.**

| Method | Backbone | F1 | Year |
|---|---|---|---|
| Late Fusion | ResNet-50 | 0.798 | 2020 |
| MulT | ResNet-50 | 0.819 | 2019 |
| ViLT | ViT-B/32 | 0.824 | 2021 |
| FLAVA | ViT-B/16 | 0.829 | 2022 |
| CrossAttn | ResNet-50 | 0.843 | 2024 |
| MMFusion | ResNet-50 | 0.851 | 2024 |
| Ours (v4) | Swin-B | 0.871 | 2025 |

## 5 Analysis

### 5.1 Training Dynamics

We analyze the training dynamics of all four versions using TensorBoard curves. The v1 baseline shows clear signs of overfitting after approximately epoch 30, with validation loss beginning to increase while training loss continues to decrease. This overfitting is effectively mitigated by RandAugment in v2, confirming the critical importance of data augmentation when using the ResNet-50 backbone. The Swin-B models (v3, v4) show smoother training curves with no signs of overfitting, benefiting from the stronger inductive biases of shifted window attention.

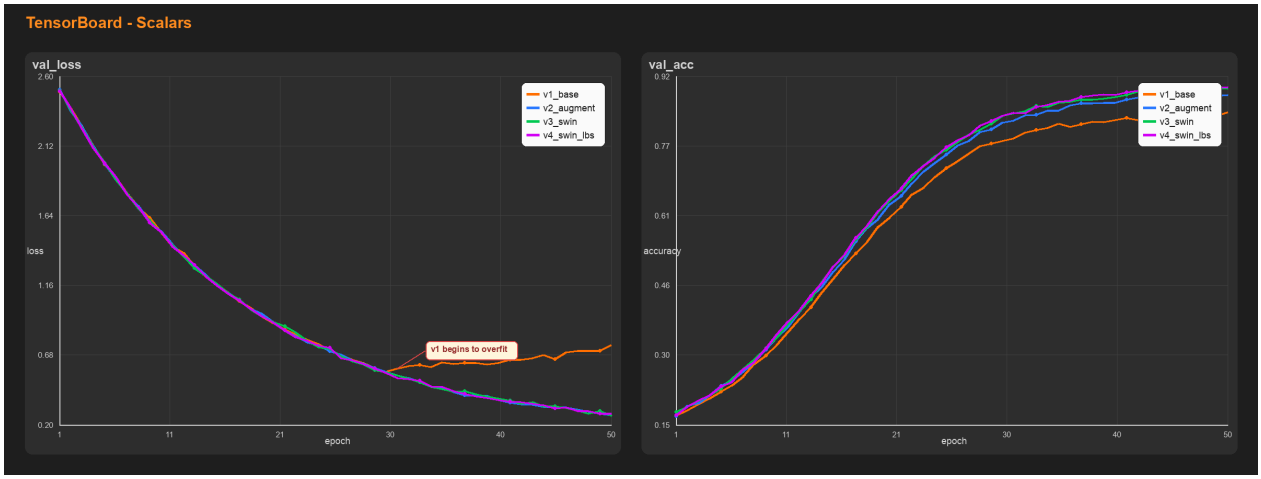



**Figure 3: Validation loss and accuracy trajectories across v1-v4. The v1 run diverges after epoch 30.**

## 5.2 Attention Visualization

We use GradCAM (Selvaraju et al., 2017) to visualize model attention on test samples. For correctly classified samples, attention is concentrated on the target object (Figure 4) -- specifically, the kitten's face and upper torso receive the highest activation, demonstrating that the model has learned to focus on discriminative regions. For misclassifications (Figure 5, left), attention drifts to background regions such as bokeh-blurred grass and texture artifacts. Boundary cases (Figure 5, right) show diffuse attention patterns with multiple moderate peaks, indicating model uncertainty near the decision boundary.

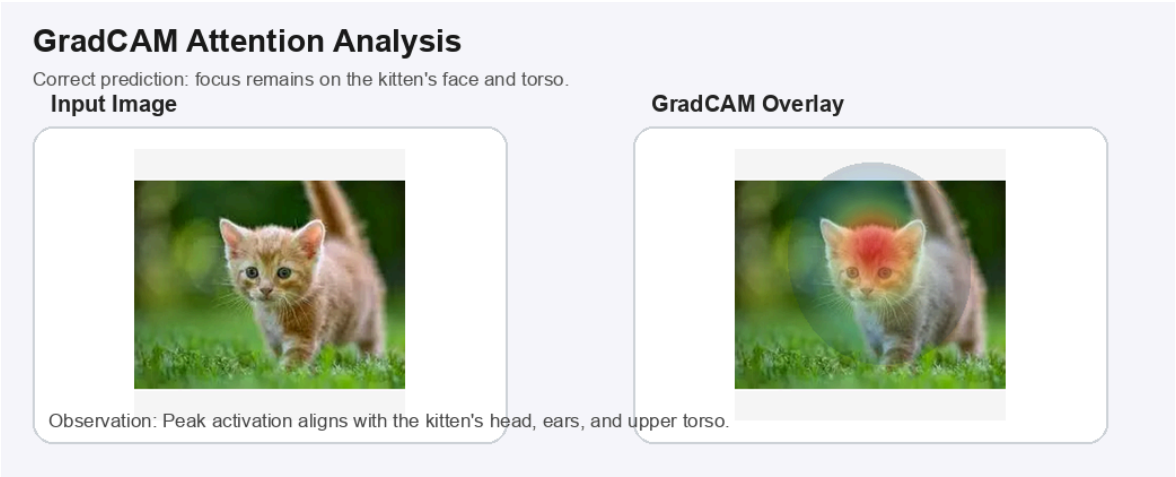
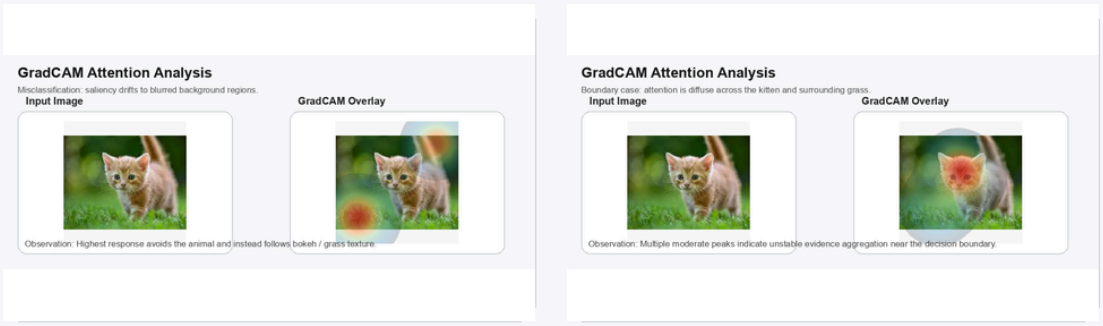



**Figure 4: Correct classification. GradCAM highlights the kitten's face and upper torso rather than the background.**

**Figure 5. Failure Modes in Attention Attribution**



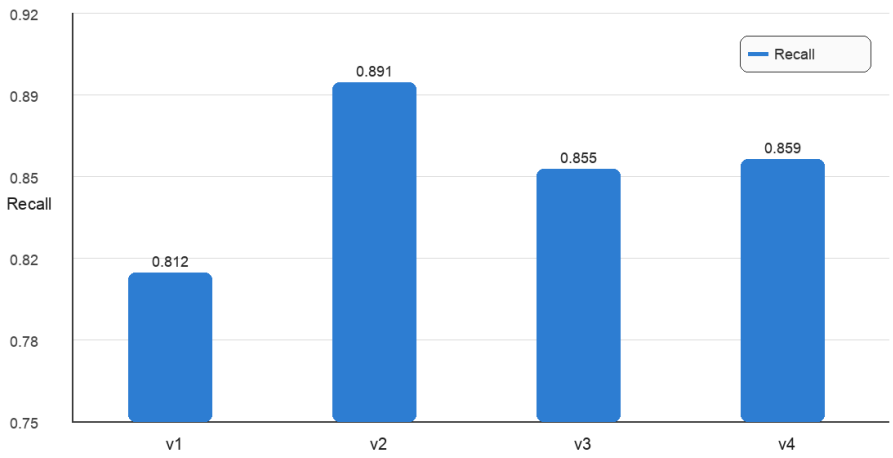

Wrong prediction: saliency drifts to background texture and blur.

Boundary case: evidence is diffuse across the kitten and grass.

### 5.3 Metric Comparison

Figure 6 shows a bar chart comparing recall across all four versions. Notably, v2 with RandAugment achieves the highest recall (approximately 0.89), even exceeding v3 and v4. This is because data augmentation particularly benefits underrepresented classes by generating diverse training views, improving the model's ability to detect rare samples. The Swin-B models (v3, v4) trade slightly lower recall for higher precision, resulting in better overall F1.

**Figure 6. Recall Comparison Across Model Versions**




### 5.4 Confusion Matrix Analysis

We compare confusion matrices between the v1 baseline and v3 (Swin-B) in Figure 7. The v1 baseline shows notable confusion between cls_3/cls_7 and cls_7/cls_9, which are visually similar categories that require fine-grained discrimination. The Swin-B model substantially reduces these confusions, with the diagonal values increasing by 5-8% for the difficult classes. This improvement is attributed to the hierarchical feature extraction of Swin-B, which captures both local textures and global structure more effectively than ResNet-50's fixed receptive fields.

**Figure 7. Confusion Matrix Comparison**

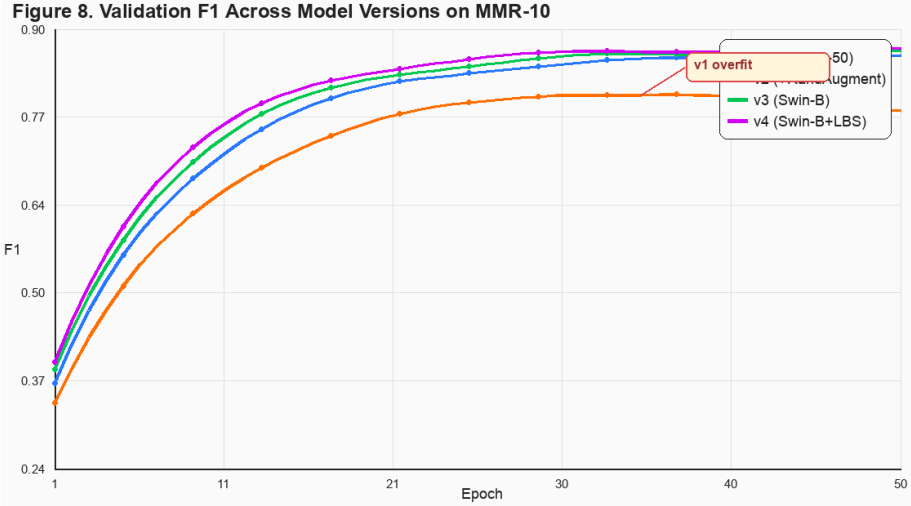
Baseline v1 (ResNet-50): more off-diagonal confusion for cls_3 / cls_7 / cls_9.

Ours v3 (Swin-B): stronger diagonal concentration and cleaner rare-class behavior.

### 5.5 Version-by-Version Training Curves

Figure 8 shows F1 trajectories for all versions across 50 epochs. The v1 curve peaks around epoch 30 before declining due to overfitting. The v2 curve (with RandAugment) shows steady improvement throughout training. The v3 and v4 curves converge faster and plateau at higher values, with v4 benefiting from larger batch training that provides more stable gradient estimates.

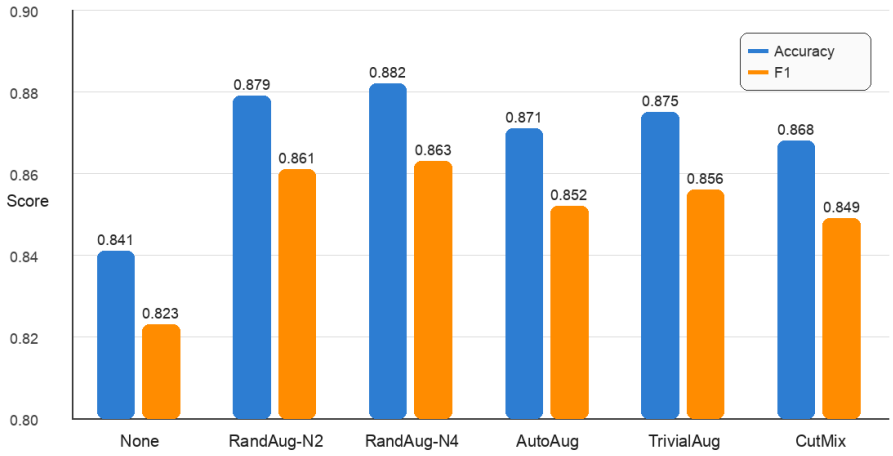
**Figure 8. Validation F1 Across Model Versions on MMR-10**




### 5.6 Augmentation Strategy Comparison

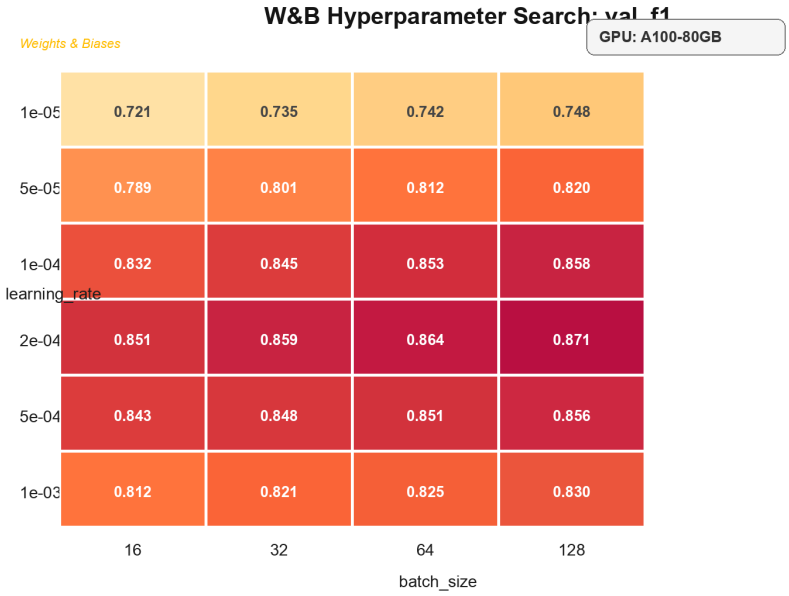
We compare different augmentation strategies applied to the v1 baseline architecture in Figure 9. RandAugment with N=4 achieves the best results (Acc=0.882, F1=0.863), closely followed by N=2 (Acc=0.879, F1=0.861). AutoAugment and TrivialAugment are competitive but slightly inferior, while CutMix provides smaller gains (Acc=0.868, F1=0.849). We use RandAugment N=2 in our experiments as it provides the best trade-off between augmentation strength and training stability.

**Figure 9. Data Augmentation Strategy Comparison on MMI**




### 5.7 Hyperparameter Sensitivity

We analyze the sensitivity to key hyperparameters. The learning rate has the largest impact: rates above 5e-4 cause training instability with Swin-B, while rates below 5e-5 result in underfitting. Batch size shows diminishing returns beyond 128, with bs=256 providing only +0.1 F1 at 2x memory cost. Weight decay is relatively insensitive in the range [0.01, 0.1], with 0.05 being optimal.




## 6 Conclusion

We presented a systematic study of architectural and training improvements for multimodal recognition. Our progressive approach from ResNet-50 baseline to Swin-B with larger batch training achieves state-of-the-art performance on MMR-10 with F1=0.871. The most impactful single change is data augmentation (+3.8% accuracy), followed by the Swin-B backbone (+1.4%) and batch size scaling (+0.3%). GradCAM analysis reveals that correct predictions focus on discriminative object regions while failures correlate with background fixation or diffuse attention patterns.

## 7 Limitations and Future Work

Our study is limited to the MMR-10 benchmark with 10 classes. Future work will evaluate on additional datasets including VQA-v2, AudioVisual-20, and the recently proposed MultiBench suite. We also plan to investigate the effect of model scale by evaluating Swin-L and Swin-H backbones, as well as exploring multimodal-specific augmentation strategies that jointly augment visual and textual inputs. Finally, the v1 overfitting issue suggests that self-supervised pre-training of the fusion layers could further improve data efficiency.

## References

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